

Web Scrapping, AI Website Intelligence & Dataset Engineering

This PDF contains the saved conversation discussing: Web scraping tools AI-powered website analysis Dataset generation for ML Website summarization CSS/style extraction Technology detection

Core Problem Statement

The objective was to build or use systems capable of: Scraping single or multiple webpages Extracting structured datasets for ML modelling Summarizing websites automatically Extracting themes, styles, fonts and colors Analyzing frameworks and technologies used

Recommended Scraping Tools

Scrapy

<https://scrapy.org>

Beautiful Soup

<https://www.crummy.com/software/BeautifulSoup/>

Playwright

<https://playwright.dev>

Selenium

<https://www.selenium.dev>

AI Extraction & Summarization Tools

Firecrawl

<https://www.firecrawl.dev>

Jina AI Reader

<https://jina.ai/reader>

Trafilatura

<https://trafilatura.readthedocs.io>

Technology & Theme Detection

BuiltWith

<https://builtwith.com>

Wappalyzer

<https://www.wappalyzer.com>

Extractable features: Fonts Primary colors Dark/light themes Animations Layout structures Framework usage

Suggested Architecture

URL INPUT → Crawl Website → Extract Text / Styles / Metadata → LLM Summary → Theme Detection → Save Structured Dataset

Potential ML Features

Domain category Website theme Framework used Readability score SEO metrics Sentiment analysis Accessibility score

Suggested Tech Stack

Backend: Python, FastAPI

Scraping: Scrapy, Playwright, BeautifulSoup

AI: OpenAI API, LangChain, LlamaIndex

Storage: PostgreSQL, MongoDB, Parquet

Frontend: Streamlit, Next.js

Roadmap

Single-page scraper Multi-page crawler Dynamic JS rendering AI summarization & classification Full analytics platform
